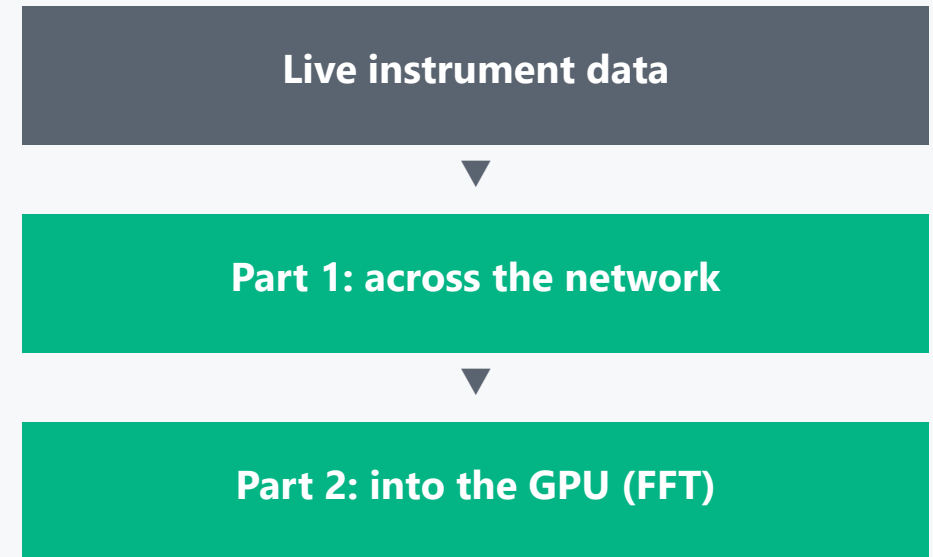


Getting Live Hardware Data to the GPU, Faster

Dami Thomas

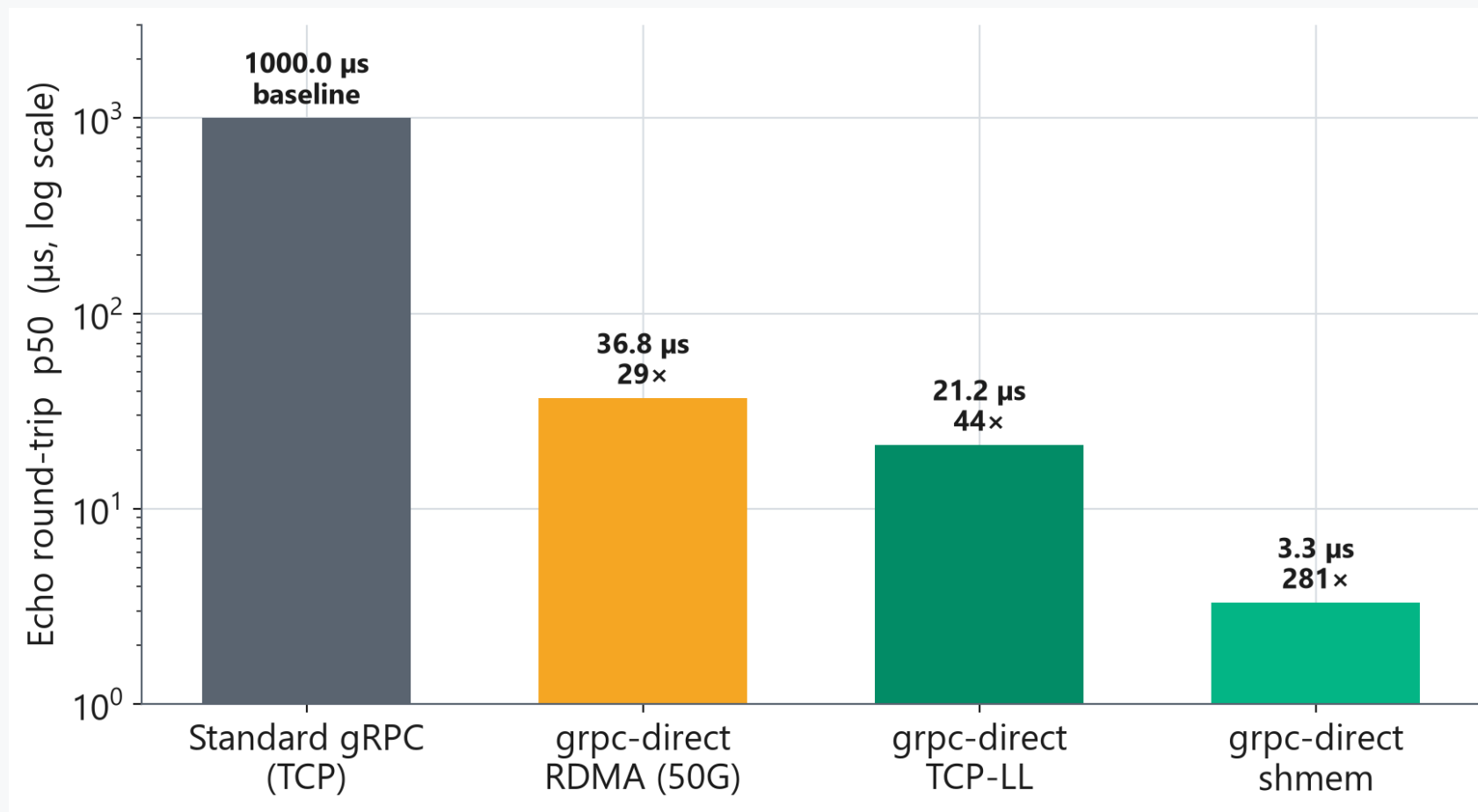
The data path, not the math, is what slows real time GPU work

- Instruments send data without pause, and it has to reach the GPU in microseconds
- The usual network path adds delay from the operating system and from copying data in memory
- That delay, not the FFT math, is what usually holds real time work back
- So we measured the path in two stages, then compared transports at each stage

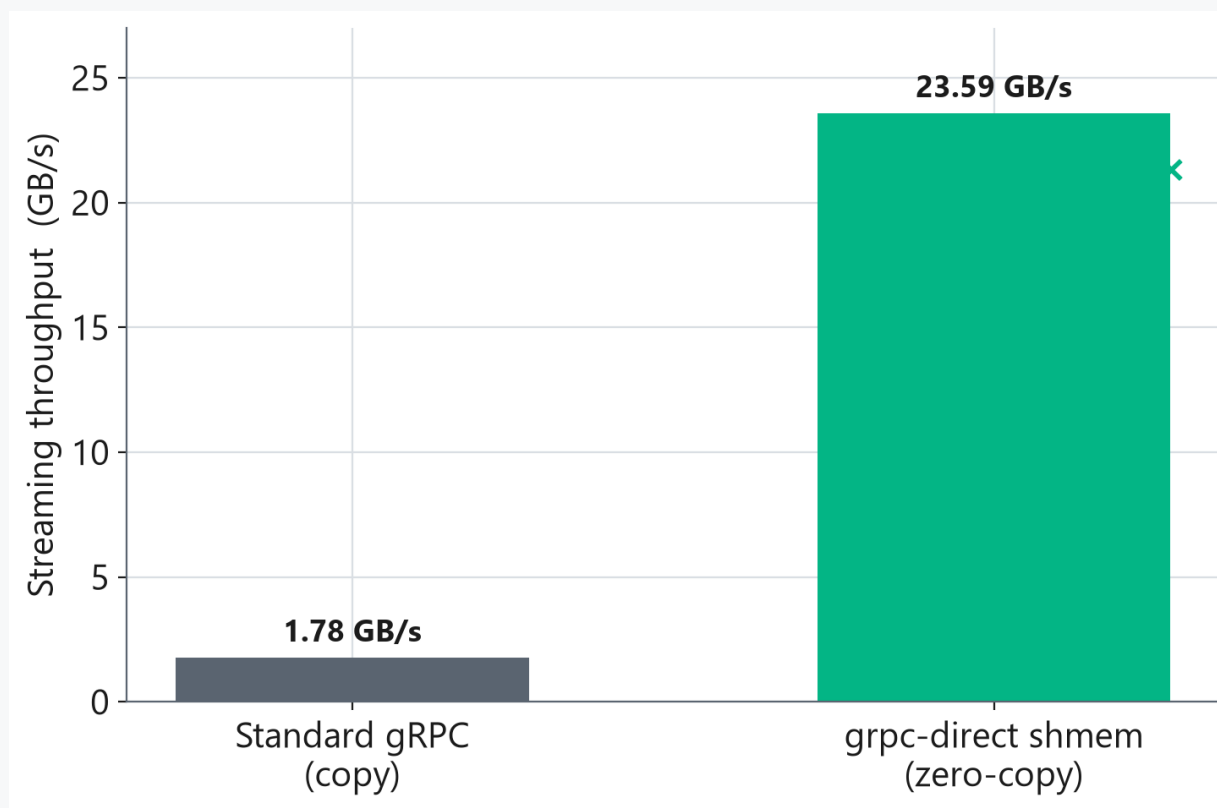


I measured the delay added at each stage.

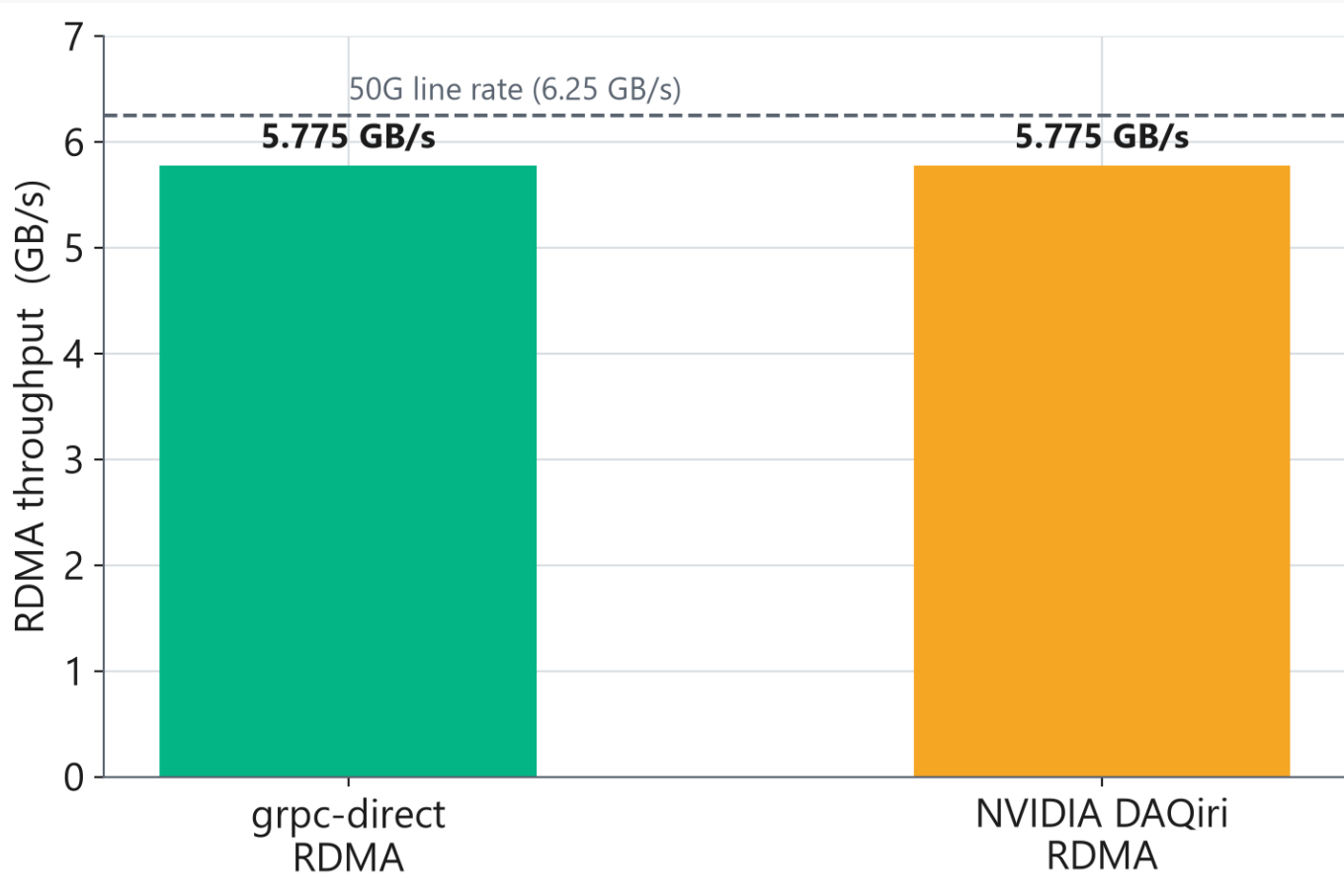
Keeping the same code but swapping the transport cut latency up to 281 times



Skipping the memory copies gave almost four times the streaming throughput

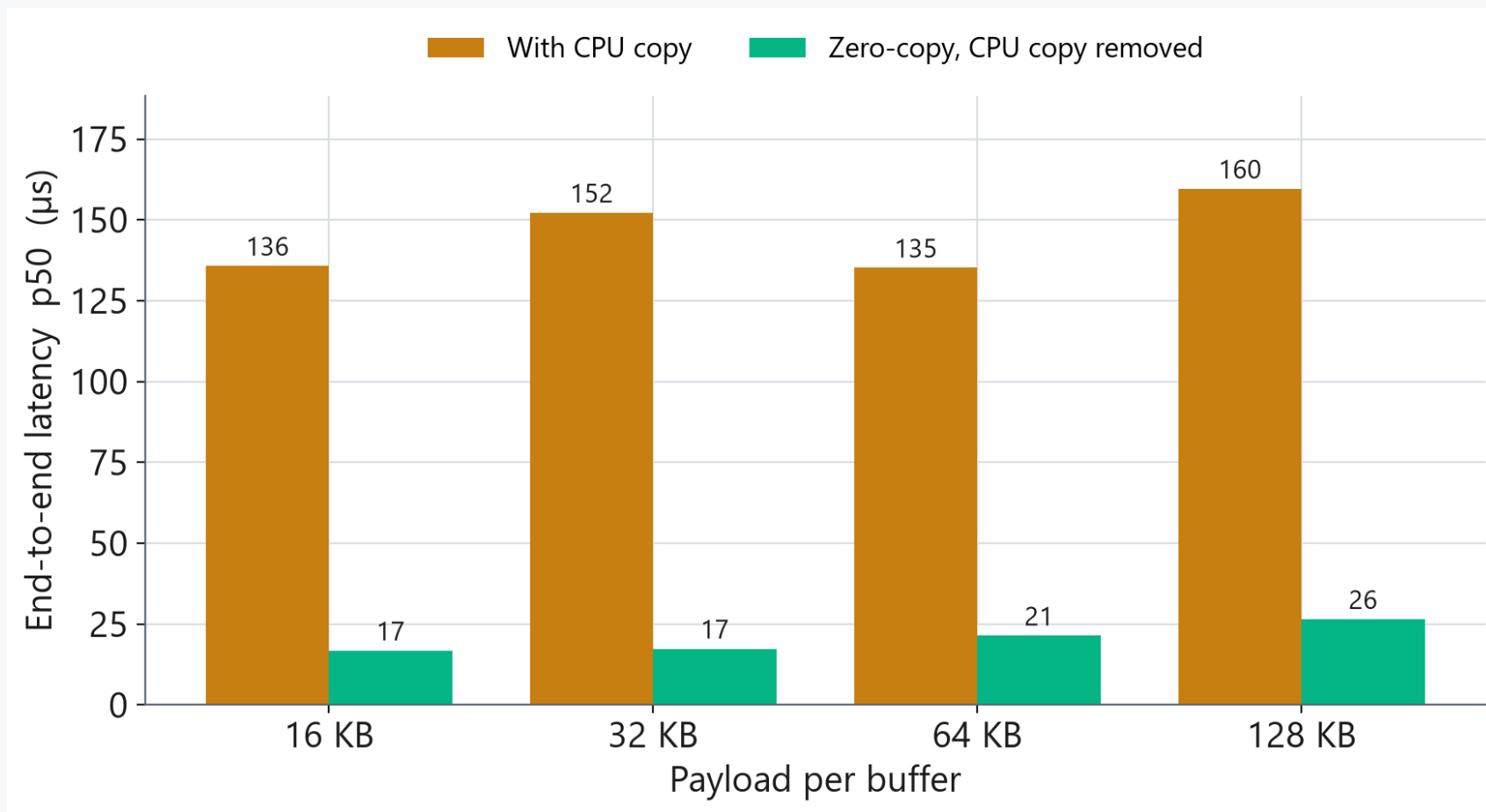


On raw network throughput, our transport tied NVIDIA's dedicated framework



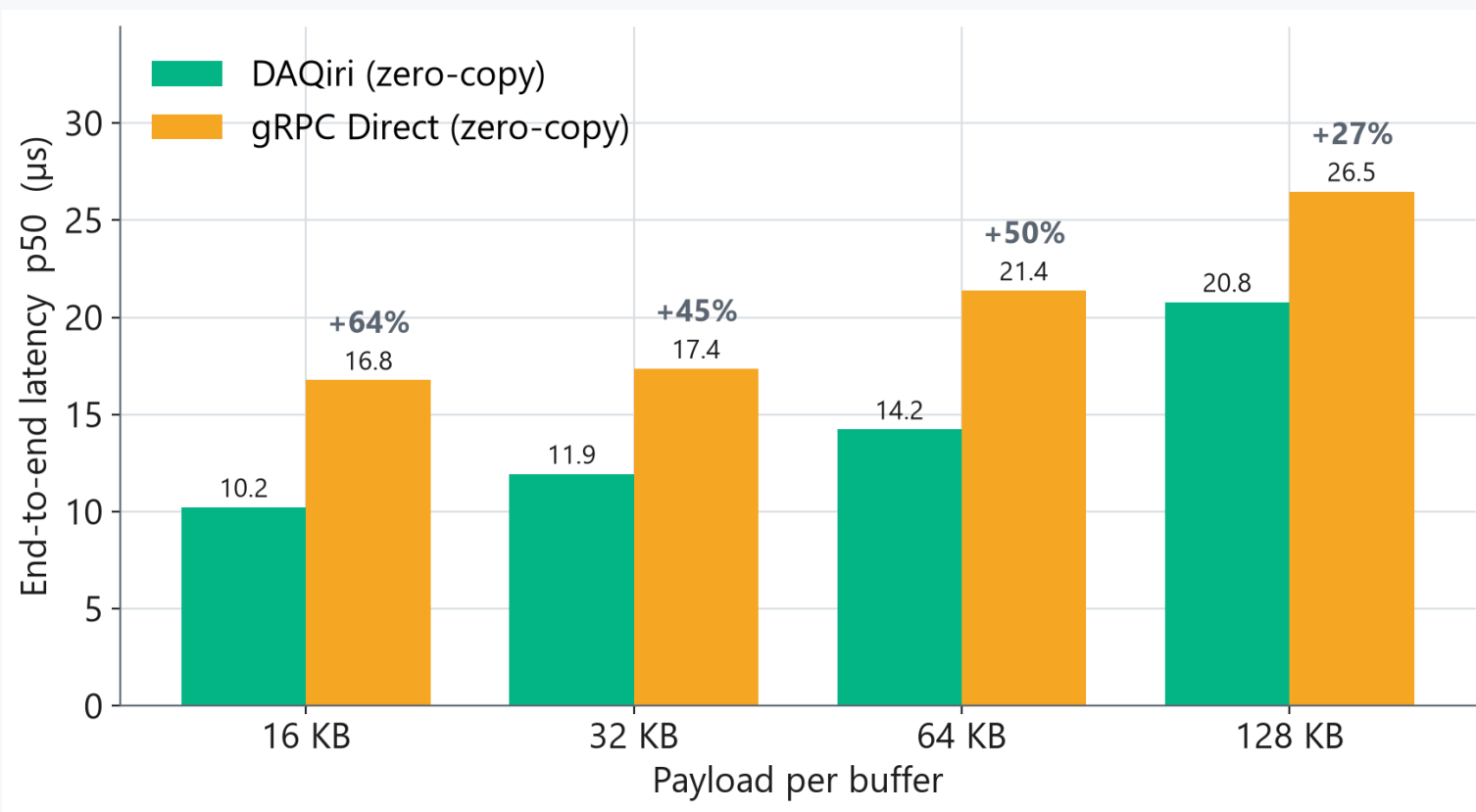
- This test measures how much data per second crosses the 50-gigabit link. Both moved about **5.8 gigabytes per second**, within one percent of each other.
- That is a throughput result. How fast a single buffer reaches the GPU is a separate question, and we answer it in Part 2.

In the GPU pipeline, both transports now skip the big CPU copy



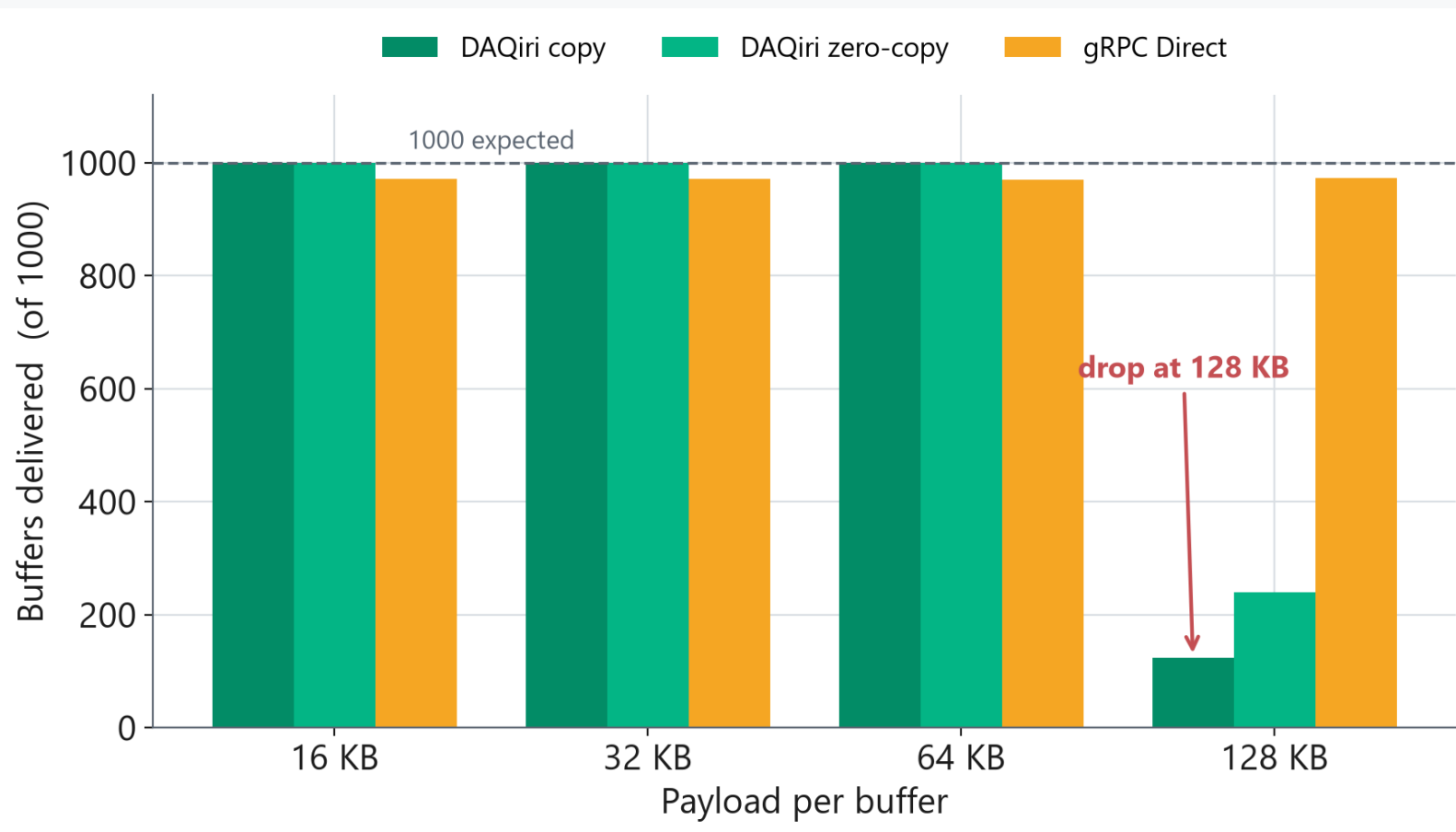
One small copy still happens on the GPU. The large copy on the CPU side is gone.

Getting a buffer into the GPU, DAQiri was faster at every size we tested



- The two transports tied on raw throughput in Part 1. For the time one buffer takes to reach the GPU, DAQiri came out ahead at every payload.
- Throughput and latency are separate questions, and here they point different ways.

One open item: at the largest payload, DAQiri delivered fewer messages



- The messages that did arrive were still fast. I am checking whether this is a limit of the test link or a setting we can change.

The bottom line: moving data faster lets the GPU act in real time

- Up to 281 times lower latency and about four times the throughput, with no change to application code
- A software transport using our gRPC Direct implementation that keeps pace with Nvidia's dedicated acquisition hardware on the network
- A GPU pipeline that reaches the accelerator in tens of microseconds instead of hundreds
- That headroom makes tighter control loops and faster inference practical, and widens the hardware we can support

Next Steps

- Find what causes the large payload delivery limit and run the test again
- Confirm the results on the main network to GPU path using real hardware data
- Write up the findings as a transport recommendation other NI teams can use

Thanks. Happy to take questions.